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1. Description of the tool

The tool developed by the UniPD unit is called pLCA-NEST (prospective Life Cycle Assessment-Neighbourhood Energy system Selection Tool). pLCA-NEST has been designed to integrate the environmental evaluation of building-wise prospective scenarios of urban energy systems (heating, cooling, appliances) and their ranking with the m-ITARA-trigonometric t-conorm and t-norm fuzzy OPA–CoCoSo-Bonferroni method.

In its LCA part this tool is strongly based on the structure of the LCA4DHC tool (Autelitano and Famiglietti, 2026) developed by the PoliMI unit of Task 8.5.2. The LCA4DHC structure and codes have been adapted to prospective LCA, a different object (from a single DHC to building by building) and different functional units.

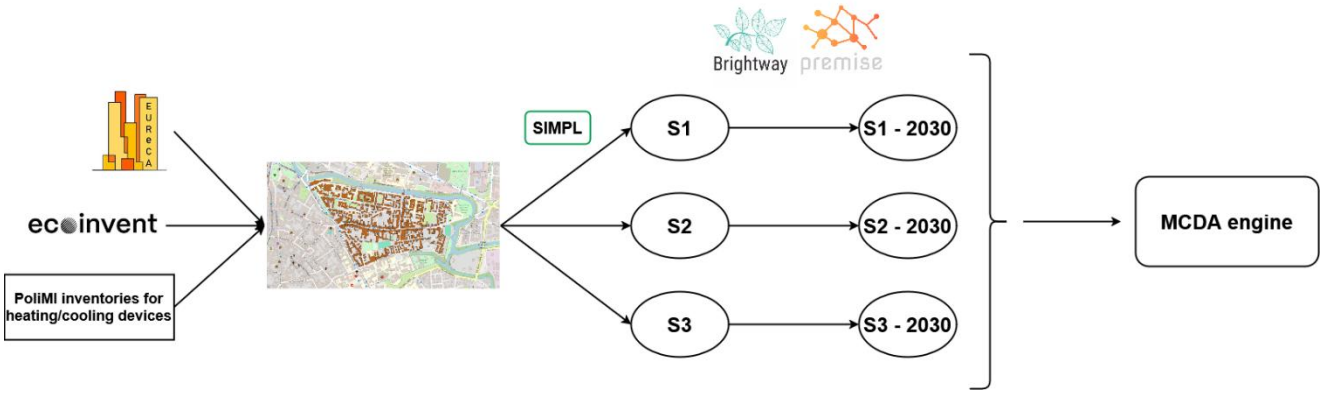


Figure 1 Logical structure of pLCA-NEST

1.1 Aims of the tool

pLCA-NEST can be used to:

- Evaluate the environmental impacts of final energy use for heating, cooling and appliances of each building of an urban stack at a certain point in the future from a life cycle perspective
- Define a ranking of different scenarios in terms of minimization of the potential environmental impacts

Moreover, since pLCA-NEST has been built on top of the LCA4DHC computational structure, it can:

- Ensure compliance with EPBD (Energy Performance of Buildings Directive, version 4) requirements

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- Support standardized and automated LCA calculations according to EN 15804 + A2 and EN 50693
- Enable assessments per technology and per life cycle module (from A to D) - from-cradle-to-cradle
- Provide a modular, transparent, and reproducible computational framework

1.2 Structure of the tool

The tool is made of:

- A worksheet input file (e.g, *DataInput_S1*) containing the general information about the scenario and the specific information for each energy system that has been selected. A blank version of this file is available in the folder *blank_inputs*;
- The worksheet *EndOfLifeInputParameters*, that stores specific end-of-life parameters for the energy systems;
- The Python package *plca*, which reads the input file and the end-of-life parameters, sets up the *Brightway* project and the background databases, builds the life cycle model of each energy system and life cycle module, performs the LCA calculations and aggregates and exports the results;
- The module *plca/ecoinvent_datasets.py*, that imports the datasets used to characterize all the material and energy flows of the energy systems from the background database (*ecoinvent* or its modified versions) and stores them as variables that can be read from the other files;
- The Python package *energy_systems*, which contains a module for each energy system that calculates the values of all the material and energy flows for the life cycle of that energy system based on the input data;
- The Jupyter notebook *RUN_pLCA*, which performs the LCA calculations for the pLCA scenario under study, by calling the functions of the plca package in order. This file imports the background database (*ecoinvent*), modifies it via *premise* and then uses the *brightway* library (Mutel, 2017) to build the trees representing each life cycle module of the energy systems

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in the scenario and to perform the LCA calculations. Then it exports the results for the scenario;

- The Python script *batch_run.py*, which executes *RUN_pLCA* unattended for several scenarios, one after the other, reading the answers to the notebook's questions from one text file per scenario;
- A worksheet input file called *scenario_summary* in which the results of all the calculated scenarios for the relevant environmental criteria (C1: Acidification, C2: Climate Change: Total, C3: Eutrophication: Freshwater, C4: Eutrophication: Marine, C5: Eutrophication: Terrestrial, C6: Ozone Depletion, C7: Total Use Of Non-Renewable Primary Energy Resources, C8: Particulate Matter Formation, C9: Photochemical Oxidant Formation: Human Health, C10: Material Resources: Metals/Minerals, C11: Water Use). An example of such file is provided with the tool.
- The Jupyter notebook *RUN_MCDA*, which performs the calculations for the Integrated m-ITARA-trigonometric t-conorm and t-norm fuzzy OPA–CoCoSo-Bonferroni method and exports the ranking results.

1.3 Workflow

The workflow of the pLCA-NEST tool is the following:

Step 1: The data input file must be filled according to the scenario to be modelled. The information necessary to run the tool can be divided into general information about the scenario and specific information about each energy system to be included in the model.

The general information about the scenario can be summed up as follows:

- Reference Study Period, default is 1 year;
- Heating and cooling load [kW] for each building;
- Heating and cooling energy demand [kWh] for each building;
- Selected energy systems for heating and cooling in each building;
- Total number of the selected energy systems for heating and cooling;
- Appliances energy demand [kWh] for each building;

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- Installed power of the PV system of each building, if present;
- Useful surface [m^2] of each building;
- An id code for each building.

Then a specific sheet is available to state the specific information about each energy system that has been selected.

A blank data input file with the necessary columns highlighted and examples of data input files are available with the tool, in the folders *blank_inputs* and *Examples*.

Step 2: Execute the *RUN_pLCA* file and follow the instructions (or use batch_run.py to run several scenarios one after the other). This file needs to be executed for each scenario, i.e. for each input file. The results are exported in worksheets in different shapes:

- A worksheet for space heating, in which all the impacts for each life cycle module of the heating-related energy systems are reported for each building (row). Full life cycle results are available too;
- A worksheet for space cooling, in which all the impacts for each life cycle module of the cooling-related energy systems are reported for each building (row). Full life cycle results are available too;
- A worksheet for appliances, in which all the impacts for each life cycle module of the appliances-related energy systems are reported for each building (row). Full life cycle results are available too;
- A worksheet for all the functions, that sums up the previous ones and reports all the impacts for each life cycle module for each building (row). Full life cycle results are available too;
- A worksheet for the whole neighbourhood, which reports the impacts of the whole life cycle, with and without module D, and of each life cycle module as area-weighted values (the total impact of the neighbourhood divided by its total useful surface), together with the whole life cycle impacts split by function (space heating, space cooling, appliances).

Step 3: Once the LCA results are calculated for all the scenarios to be modelled, the whole neighbourhood impacts need to be collected in the *scenario_summary* file, taking for every scenario the same sheet of the whole neighbourhood results: Whole Lifecycle (hood), or Whole Lifecycle (no D, hood) to leave out module D.

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Step 4: Execute the *RUN_MCDA* file. The results of the ranking according to the Integrated m-ITARA-trigonometric t-conorm and t-norm fuzzy OPA–Fuzzy CoCoSo Bonferroni method are exported to the file *integrated_mcda_results*. In this file all the results for all the steps of the MCDA method are reported. To facilitate the user, the results are exported in a short form as *integrated_mcda_results_short*, in which only the final weights of the criteria and the results are reported.

1.4 Requirements

The pLCA-NEST tool requires:

- Username and password of an active license for the *ecoinvent* database (the tool has been built on *ecoinvent v3.11 cut-off*; other versions can be used but make sure to verify if any change in the names of the dataset has occurred)
- The *premise* encryption key from the developers of *premise*

The Python packages listed in the file *requirements.txt*, in the versions with which the tool has been developed and tested (Python 3.12).

1.5 Disclaimer

The pLCA-NEST tool is provided for research, academic, and technical support purposes only.

- The authors do not guarantee the absence of errors, inaccuracies, or omissions.
- The authors accept no liability for any direct or indirect damages resulting from the use of the tool.
- Users are fully responsible for validating the correctness, robustness, and suitability of the results for their specific application.
- Results should be independently reviewed, especially when used for regulatory, policy, or decision-making purposes.
- The input file must be fully completed in all its sections, with all required data fields properly filled in. Incomplete or partially compiled input files may lead to invalid or misleading results.

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1.6 Status

The tool is:

- **Stable** – Core computational workflow is fully implemented and functional
- **Experimental** – Scientific and methodological validation is ongoing
- **Under active development** – Continuous updates and improvements could be delivered

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